

CONTACT INFORMATION	Chicago, Illinois, USA  Portfolio,  Scholar  Primary : rajeshr12@zohomail.com , University : rrame12@uic.edu  https://www.linkedin.com/in/rajeshrrajeshr/  https://github.com/its-rajesh
RESEARCH INTERESTS	Audio signal processing and machine learning, with emphasis on source separation, spatial and multichannel audio, interference reduction, distance-aware acoustic modeling, distributed acoustic sensor networks, and learning-based signal representations.
EDUCATION	University of Illinois Chicago Aug 2024 - Current PhD in Audio Signal Processing at Listening Technology Lab — Area : Audio Signal Processing — CGPA : 4/4 (Current) — Advisor : Prof. Ryan M. Corey  — Selected Courses : <i>Digital Signal Processing II, Adaptive Digital Filters, Statistical Digital Signal Processing, Audio Signal Processing, Information and Learning, Random Signal Analysis, Estimation and Detection, and Elements of Information Theory.</i> Indian Institute of Technology Mandi (IIT Mandi) Aug 2021 - April 2024 MS by Research at the School of Computing and Electrical Engineering — Area : Audio Signal Processing - Music Source Separation and Music Bleeding Removal — CGPA : 8.73/10 — Dissertation : <i>Music Source Separation</i> — Advisor : Prof. Padmanabhan Rajan  — Selected Courses : <i>Deep Learning and Applications, Matrix Theory, Applied Optimization, Probability and Random Processes, Pattern Recognition, Programming Practicum for Signal Processing, Digital Image Processing</i> Anna University - S.A. Engineering College Aug 2014 - May 2018 Bachelors of Engineering - Electronics and Communication Engineering (ECE) — University Rank (32nd out of 17,534), 2nd in the institute — Highest Honours (top 0.18%), CGPA : 8.9/10 — Top Courses : <i>Digital Signal Processing, Signals and Systems, Linear Algebra, Transforms and Differential Equations, Multivariate Calculus and Complex Integration, Probability and Random Processes</i>
RESEARCH EXPERIENCE	Distance-Aware Source Separation Aug 2025 – Present Developing multichannel source-separation methods that explicitly exploit source-to-microphone distance to distinguish and reconstruct near and far sources across speech, music, and environmental audio. ★ Designed distance-aware spatial representations using ILD, IPD, ITD, multi-resolution energy features, and DRR-based reverberation cues. ★ Investigating neural conditioning with learned distance embeddings for robust separation across varying acoustic environments. ★ Project page :  Distributed Acoustic Sensor Networks Aug 2025 – Present Developing cooperative speech-enhancement and source-separation methods for distributed acoustic sensor networks under communication and resource constraints. ★ Investigating bandwidth-constrained information sharing to approach centralized enhancement performance while reducing inter-node communication. ★ Developed and evaluated DANSE-style distributed beamforming and neural message-passing approaches under communication and synchronization constraints.

- ★ Studying node-specific enhancement, source localization, and spatial inference in distributed microphone networks.

• **Interference Reduction**

Aug 2024 – Sep 2025

Developing learning-based and signal-processing methods for microphone-bleed reduction in highly reverberant multichannel recordings.

- ★ Investigated generative, waveform, STFT, and wavelet-domain architectures for interference suppression while preserving target-source fidelity.
- ★ Developed learnable wavelet front ends for end-to-end interference reduction and analyzed their learned representations relative to fixed transforms.
- ★ Evaluated methods using simulated acoustics, measured room impulse responses, recorded noise, and cross-domain musical recordings.
- ★ Project page : 

• **Music Source Separation & Interference Reduction**

Dec 2021 – Apr 2024

Developed deep-learning and signal-processing methods for music source separation and microphone-bleed reduction, with emphasis on Indian classical music recordings.

- ★ Investigated source separation beyond conventional 4–5 stem formulations for Carnatic and Hindustani music involving a larger number of interacting sources.
- ★ Developed Wave-U-Net architectures augmented with graph attention mechanisms to model interactions among multiple sources and microphone channels.
- ★ Explored learning-free optimization, convolutional autoencoders, generative models, and graph neural networks for interference reduction.
- ★ Project pages :  

SELECTED PUBLICATIONS

- [C6] **Rajesh R**, Rashan Fernando, Yuezhong Xu, and Ryan M. Corey, “Cooperative Node-Specific Speech Enhancement in Resource-Constrained Acoustic Sensor Networks.” 2026 14th International Workshop on Acoustic Signal Enhancement (IWAENC). IEEE, 2026  
- [C5] Rashan Fernando, **Rajesh R**, Yuezhong Xu, and Ryan M. Corey, “Geometry-driven signal alignment and spatialization for assistive listening with a remote microphone array.” 2026 14th International Workshop on Acoustic Signal Enhancement (IWAENC). IEEE, 2026
- [C4] **Rajesh R**, Rashan Fernando, Padmanabhan Rajan, and Ryan M. Corey. “Bleed No More : Generative Interference Reduction for Musical Recordings.” 2026 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2026.  
- [C3] **Rajesh R**, Rashan Fernando, and Ryan M. Corey, “Distance-Aware Source Separation via Multi-Microphone Spatial Features in Reverberant Rooms,” *Proceedings of Meetings on Acoustics (POMA)*, 2026. **Winner, POMA Student Paper Competition (ASA 2026)**.  
- [C2] Rashan Fernando, **Rajesh R**, Kanad Sarkar, Manan Mittal, Andrew Singer, and Ryan M. Corey. ‘Room impulse response interpolation using a time-warping neural network’ Proceedings of Meetings on Acoustics (POMA), 2026.
- [C1] **Rajesh R** and Padmanabhan Rajan, “Neural Networks for Interference Reduction in Multi-Track Recordings,” 2023 IEEE Workshop on Applications of Signal Processing to Audio and Acoustics (WASPAA), New Paltz, NY, USA, 2023, pp. 1-5.  

IN REVIEW & PRE-PRINTS

- [J2] **Rajesh R**, Rashan Fernando, and Ryan M. Corey. “End-to-End Learnable DWT Frontends for Multitrack Music Bleed Reduction.”
- [J1] **Rajesh R** and Padmanabhan Rajan. “GIRNet : A Graph-Based Interference Reduction in Live Microphone Recordings.”  

PROFESSIONAL EXPERIENCE

- **Research Intern** Nov 2023 – Jan 2024
Dubverse.ai, India
Conducted research on audio source separation.
- **Data Science Engineer** Jan 2018 – May 2019
DeTect Technologies Pvt. Ltd., IIT Madras Research Park, Chennai
Developed an end-to-end automated report-generation pipeline and PyQt-based software tools for internal engineering applications.
 - Tools : *PyQt5*, *Pandas*, *Matplotlib*, *OpenCV*, *Git*

TEACHING ASSISTANTSHIPS	• EC317 Digital Signal Processing, University of Illinois Chicago	Spring 2025
	• EE511 Computer Vision, IIT Mandi	Fall 2023
	• CS671 Deep Learning & Applications, IIT Mandi	Spring 2023
	• IC151 Computing and Data Science, IIT Mandi	Fall 2022
	• CS571 Programming Practicum (Signal Processing), IIT Mandi	Fall 2022
AWARDS & HONORS	— POMA Student Paper Competition Winner , Acoustical Society of America, 2026.	
	— University Rank – B.E. : Ranked 32nd among 17,534 students (top 0.18%).	
	— Academic Excellence Award – B.E. : Received the academic excellence award in every semester.	
ACADEMIC SERVICE & WORKSHOPS	— Volunteer, 23rd International Society for Music Information Retrieval Conference (ISMIR 2022), Indian Institute of Science, Bengaluru.	
	— CompMusic Workshop on Computational Musicology and Carnatic Music. Project : <i>Tonic Identification in Carnatic Music</i> 	
	— Winter School on Speech and Audio Processing (WiSSAP 2023), “Music for AI,” IIT Kanpur.	
	— 25th International Conference on Speech and Computer (SPECOM 2023), Hubli and Goa, India.	
TECHNICAL SKILLS	Programming : Python, MATLAB, Bash, $\text{\LaTeX}$ Machine Learning : PyTorch, TensorFlow/Keras Audio & Signal Processing : STFT, Wavelets/DWT, Source Separation, Spatial Audio, Microphone Arrays	Scientific Computing : NumPy, SciPy, Pandas, Matplotlib, scikit-learn, Librosa, Pyroomacoustics Systems & Tools : Linux, CUDA/GPU Computing, Jupyter, PyQt Version Control & DevOps : Git, GitHub, GitHub Actions
LANGUAGES	Tamil (Native), English (Fluent)	
REFERENCES AVAILABLE UPON REQUEST	Ryan M. Corey Padmanabhan Rajan	Assistant Professor, UIC  Associate Professor, IIT Mandi 